

Prediction–Attribution Conjugacy in Adaptive Decision Problems

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October 2026

Abstract

Many consequential decisions involve uncertainty not only about what will happen, but about which parts or relations within a system matter for action. We examine a class of adaptive decision problems in which decision makers form structural accounts, make prospective predictions, act, and learn from subsequent outcomes. Within some of these problems, attribution and prediction become reciprocally linked. We describe this relationship as prediction–attribution conjugacy.

The framework separates the conjugate core from the responsive dynamics that may surround it. This perspective helps distinguish ordinary model failure from structural change and from success-induced signal decay, and it clarifies why predictive stability under routine conditions need not validate the underlying attribution. Examples from finance, clinical trials, and public policy illustrate the framework and its limits.

Keywords: prediction; attribution; adaptive decision problems; decision making; model monitoring; identification

1. Introduction

A portfolio risk manager expects the S&P 500 to remain heavily concentrated in a small number of large technology companies. The forecast matters, but it does not by itself tell the manager what to do.

An effective response depends on what lies behind the concentration. If the main concern is market-cap weighting, one strategy may be appropriate. If several large companies share exposure to the same underlying economic factor, another may be needed. If concentration reflects persistent differences in earnings growth rather than a common source of vulnerability, mechanically reducing the largest positions may create a different problem.

The manager therefore needs more than a prediction of what the market will look like. The manager also needs an account of which parts of the system matter for the decision.

This combination is common in consequential decision making. A physician may need to predict whether a patient will deteriorate and understand enough about the source of the risk to choose a treatment. A policymaker may need to anticipate the effects of an intervention and understand why similar interventions have worked differently across populations or periods. An investor may identify a profitable regularity but also need to understand why the opportunity exists in order to judge whether it is likely to survive exploitation.

The same structure can appear in ordinary life. Career or education choices are made under incomplete self-knowledge, and the choices themselves can change later opportunities and preferences.

These are not simply prediction problems followed by separate requests for explanation. Prediction and attribution can affect one another. An attribution can change what should be predicted, which conditions matter, and whether a historical predictive relationship should be expected to persist. Conversely, if

competing attributions imply different future outcomes, what subsequently happens can provide evidence about which attribution remains credible.

This paper examines that reciprocal relationship.

This paper concerns consequential decision problems in which uncertainty extends beyond outcome probabilities to the structure that matters for action. The decision maker may consider alternative accounts of that structure, choose among actions whose consequences depend on it, observe subsequent outcomes, and revise what is believed. The system may also change or respond while this learning occurs. Membership in this broad class does not require that the competing accounts be empirically distinguishable.

Within some of these problems, prediction and attribution become linked in a stronger way. We describe that relationship as prediction–attribution conjugacy.

The term does not imply that prediction and attribution are exact inverses or that one must improve at the expense of the other. Conjugacy is an additional property of the relation between them, not part of the definition of the problem class itself. Within this class, structural conjugacy is often only a screening condition: the substantive empirical question is whether feasible diagnostic variation makes the relation identified before the relevant structure changes.

The framework becomes especially consequential when knowledge is used for action. Acting on a prediction and its supporting attribution can change the system from which both were learned. Better understanding may improve prediction, but successful intervention can also weaken the very predictive relationship that made the intervention possible. A model can therefore become less accurate not only because the analysis failed, but sometimes because the decision succeeded.

The purpose of the framework is not to replace causal inference, adaptive control, experimental design, decision theory, or other approaches. It offers an alternative organizing view. Its value depends on whether that view makes important problems easier to understand, distinguishes cases that should be treated differently, and improves the questions we ask about prediction, intervention, monitoring, and learning.

2. The Adaptive Decision Problem

The framework begins with the decision problem rather than with prediction or attribution in the abstract.

A decision maker has an objective and faces an uncertain system. Only part of that system may matter for the decision, and uncertainty may concern the relevant structure as well as the eventual outcome.

We call the portion or representation of the system that matters for the objective the decision-relevant system, denoted S_D . Appendix A gives the formal notation. The idea is simple: decision makers generally do not need to understand the whole world; they need to understand enough of the part that matters for the decision they face.

Consider again the stock market. A risk manager concerned with concentration risk and a hedge fund manager searching for arbitrage opportunities may observe the same S&P 500. But they need not organize what they observe in the same way.

For the risk manager, relevant distinctions may concern common exposures, potential drawdowns, liquidity, correlations, and the contribution of individual positions to portfolio loss. For the arbitrageur, the relevant distinctions may instead concern mispricing, convergence mechanisms, execution costs, crowding, and the expected persistence of a trading opportunity.

Neither representation is necessarily a more complete account of the market. They answer different decision questions.

This decision-relative view matters for attribution. Asking simply, “Why did the market move?” admits an enormous range of possible answers. Asking, “Which parts of the market structure account for the prospective loss that this portfolio is trying to manage?” defines a much more tractable problem.

Prediction is likewise conditioned by the decision. A risk manager who attributes portfolio vulnerability to a common semiconductor exposure may need to predict losses under a semiconductor shock rather than merely forecast whether market concentration will remain high.

The decision objective therefore helps determine both what must be understood and what must be predicted. At this stage, however, the paper does not assume that alternative structural accounts are predictively distinguishable; that is part of the later conjugacy test.

$$S_{D,t} \rightarrow \text{decision} \rightarrow S_{D,t+1} \rightarrow Y_{t+1}$$

Read this as: the decision-relevant system at time t informs a decision; the decision and external events contribute to the system faced at $t + 1$; and an outcome is then observed. Two different dynamics should be distinguished. The decision maker can learn by updating beliefs and actions as evidence arrives; call this the learning loop. Separately, the system itself may be responsive, meaning that action changes the relevant outcome distribution or structure, whether mechanically or because other actors react. Prediction–attribution conjugacy can occur within the learning loop; endogenous validity loss and signal attenuation require the responsive embedding.

3. Prediction and Attribution Within the Problem

Within an adaptive decision problem, prediction and attribution answer different questions. The distinction between explanatory and predictive modeling is well established, including the fact that strong performance on one objective does not imply strong performance on the other [1,2].

Prediction asks: Given the relevant system and a contemplated condition or action, what should we expect to happen?

Attribution asks: Which parts, relations, or mechanisms within the relevant system account for differences in the outcomes that matter to the decision?

In this paper, attribution is functional rather than field-specific: it is a candidate account of which components, relations, or mechanisms within the decision-relevant system matter to consequential differences in outcomes. Attribution may be causal, diagnostic, or risk-oriented. It need not already be

prospectively distinguishable or empirically testable; whether it acquires those properties through its relation to prediction is part of the conjugacy question.

Suppose a risk manager expects concentration in a few large technology companies to persist. That is a prediction about the future state of the market. But the manager still needs to decide what, if anything, should be changed.

One attribution might be that the portfolio is exposed mainly to market-cap concentration. Another might be that several large companies share exposure to the same economic factor. A third might be that the apparent concentration is less important than it looks because the underlying businesses generate sufficiently different cash flows and risks.

These are not merely different stories about the same forecast. They can matter in two distinct ways.

First, attribution can guide problem formulation by changing what the decision maker should predict next.

$$A_D \rightarrow \text{formulation of } P_D$$

For example, a risk manager may shift from predicting whether concentration persists to predicting loss under a common-factor shock. This is attribution-guided problem formulation—that is, the attribution changes what the decision maker should predict next. It is useful, but it is not by itself the conjugacy criterion.

Second, once a prediction target and relevant action or condition are fixed, competing attributions may imply different prospective outcome distributions. If the portfolio behaves as one attribution predicted and not as another predicted under a diagnostic stress, the result can change the evidentiary balance among those accounts.

$$A_i \rightarrow P(Y | u, A_i) \rightarrow \text{evidence about } A_i$$

Here A_i denotes a particular hypothesis about the structure of S_D , and u denotes a contemplated or implemented action or diagnostic condition. Thus $P(Y | u, A_i)$ is the prediction conditional on that structural hypothesis. Where causal semantics are warranted, u should be read interventionally rather than as ordinary observational conditioning.

Prediction still does not contain attribution, and predictive evidence need not uniquely identify an attribution. Its diagnostic value depends on whether the observed condition actually separates the competing accounts and whether selection, confounding, or structural change can be ruled out sufficiently for the task.

The conjugacy claim developed next concerns this second, diagnostically reciprocal relation. Attribution-guided problem formulation remains an important consequence of taking attribution seriously, but it should not be confused with the fixed-target relation used to test conjugacy.

4. Prediction–Attribution Conjugacy

For a given decision task, prediction and attribution are conjugate when at least some decision-relevant attributional differences imply distinguishable prospective outcomes under a common informative condition, and observations under such conditions can in turn revise the relative credibility of those attributions.

$$A_D \leftrightarrow P_D$$

The term conjugate has several precise meanings across mathematics and the sciences, but its ordinary relational sense is simpler: two things are paired or coupled through a defined relation. We use conjugacy in that limited organizing sense here. The two inquiries remain distinct, but attribution can give prediction diagnostic structure while prediction can give attribution empirical exposure. No specific pre-existing mathematical conjugacy is assumed.

The conjugate core should be distinguished from the surrounding learning loop and, when present, the responsive embedding. The core is the diagnostic relation $A_D \leftrightarrow P_D$. A decision maker may update this pair even in a largely stationary system. When the pair informs action in a responsive system, the larger structure becomes $A_D \leftrightarrow P_D \rightarrow D \rightarrow S'$. The stronger claims about endogenous validity loss and signal attenuation arise from that responsive embedding, not from conjugacy alone.

The relation is reciprocal but not symmetric. Alternative attributions generate prospective implications; predictive observations can reweight those attributions, but prediction does not logically contain attribution and attribution need not produce precise forecasts in every setting.

Suppose both A_1 and A_2 are plausible given current evidence, but they imply different responses to some relevant action u :

$$P(Y | u, A_1) \neq P(Y | u, A_2)$$

The attributional difference now has prospective meaning. If later outcomes are observed under conditions that actually distinguish the accounts—and the relevant structure has not changed so much that the comparison is invalid—the evidentiary balance can shift. Bayesian model discrimination provides a neighboring formal precedent for using prospective outcomes to distinguish competing accounts [14].

This makes the relation different from post-hoc explanation. A post-hoc account can fit an outcome without changing expectations about what would happen under another condition. Such an explanation may be useful, but it does not participate in the conjugate structure described here.

It is useful to distinguish structural from identified conjugacy. Structural conjugacy exists when decision-relevant attributions imply different prospective outcomes under at least one common informative condition. In this problem class, that can be a relatively low bar because structural uncertainty matters precisely when alternative accounts could matter for action. Identified conjugacy carries the stronger empirical burden: feasible observations or interventions must actually provide enough diagnostic variation to discriminate among the accounts. For example, two accounts of portfolio risk may imply different stress responses in principle, yet remain unidentified if the portfolio has never encountered or been tested against a sufficiently diagnostic stress.

An in-class problem can therefore remain non-identified. A one-off policy or investment decision may depend on competing structural accounts that imply different consequences, yet offer no feasible diagnostic condition before the decision must be made. Structural conjugacy may be present while identified conjugacy is absent. The next section asks what becomes easier to understand once the relation is embedded in real decisions and, in some cases, responsive systems.

Table 1. Four related layers in the framework

Layer	Question	Role
Attribution	What structure matters?	Candidate account.
Attribution-guided formulation	What should be predicted next?	Changes target, condition, or horizon.
Attribution-conditioned prediction	What does each account predict under the same diagnostic condition?	Creates prospective differences.
Prediction–attribution conjugacy	Can those differences be observed and revise the accounts?	Reciprocal diagnostic relation.

5. What the Conjugacy View Helps Us See

5.1 Attribution can change the prediction problem

Attribution can change the prediction problem before any formal conjugacy test is possible. A risk manager who identifies a common factor may need to predict loss under a factor shock rather than merely whether concentration persists.

Causal-prediction research similarly emphasizes that ordinary predictions may be inadequate when the relevant question concerns outcomes under hypothetical interventions [6,7]. This formulation role should be distinguished from attribution-conditioned prediction of a fixed outcome under a fixed diagnostic condition.

The important point is not that attribution always improves predictive accuracy. It can change which outcome, condition, horizon, or intervention should be predicted in the first place.

5.2 Predictive outcomes can discipline attribution

Attribution can easily become retrospective. Conjugacy gives it prospective exposure only when competing accounts imply different outcomes under a diagnostic condition and those outcomes can be observed with enough control over selection, confounding, and structural change to be informative.

$$A_i \rightarrow P(Y | u, A_i) \rightarrow \text{diagnostic evidence about } A_i$$

Prediction is therefore not only an output of understanding. Under suitably diagnostic conditions, it can become one source of evidence about which attribution remains credible. The word discipline is used in this limited sense, not to imply unique causal identification from ordinary forecast success.

5.3 When a responsive system changes, use can change validity

$$A_t \leftrightarrow P_t \rightarrow D_t \rightarrow S_{t+1}$$

This consequence belongs to the responsive embedding rather than to conjugacy itself. A responsive system is one in which action materially changes the relevant outcome distribution or attributional structure, whether through mechanical effects, treatment, strategic reaction, or other feedback. Once the pair informs action in such a system, the next-round structure may differ from the one on which the earlier attribution and prediction were based. Dual-control research provides a neighboring example in which action both controls a system and generates information about its uncertainty [8]. Performative prediction similarly formalizes settings in which predictions support decisions that alter future outcome distributions [9].

Conjugacy contributes by helping interpret what has changed: whether the predictive relation failed, the attributional structure changed, or an intervention successfully changed the distribution or relation on which the old predictor depended.

5.4 Successful intervention can make an old predictor worse

A particularly useful diagnostic case occurs when attribution identifies the source of a predictive signal and a responsive intervention successfully acts on that source.

Suppose X contributes materially to outcome Y , variation in X helps predict Y , and attribution identifies X as actionable. If intervention compresses relevant variation in X or changes the relation between X and Y , then the original signal may weaken:

$$\text{Signal}(X, Y)_{\text{post}} < \text{Signal}(X, Y)_{\text{pre}}$$

The old predictor may therefore become less accurate even though the intervention succeeded. This is not inevitable: a uniform shift in X or an intervention that leaves the X – Y relation intact need not attenuate the signal. The result is conditional, but its diagnostic implication is important.

A deteriorating predictor can therefore have at least three readings: the original relation was spurious, the relevant structure changed for other reasons, or the intervention itself weakened the signal. Monitoring should be designed to distinguish these possibilities rather than treating all decay as model failure. Appendix A.12 provides a simple numerical illustration.

5.5 The identification race

The framework also suggests a way to think about why understanding stabilizes in some domains and remains provisional in others.

Let λ_I denote the conceptual rate at which evidence discriminates among competing decision-relevant attributions, and let λ_R denote the rate at which the relevant attributional structure changes enough to invalidate the comparison—for example, a change in the signal–outcome relation rather than an environmental change that leaves the diagnostic contrast intact.

These are conceptual rates, not universal measured quantities. λ_I can depend on the decision maker’s own exploratory actions, while λ_R can be partly endogenous when the system responds to intervention.

$$\lambda_I \gg \lambda_R \rightarrow \text{relatively stable cumulative understanding}$$

$$\lambda_I \approx \lambda_R \rightarrow \text{persistently provisional understanding}$$

$$\lambda_I \ll \lambda_R \rightarrow \text{relevant structure changes faster than attribution can be resolved}$$

The comparison is therefore heuristic rather than a general law. It directs attention to the relative pace of diagnostic learning and structural change, while leaving room for other sources of difficulty such as noise, measurement limits, and experimental access.

6. The Structure Across Different Problems

The cases below are demonstrations rather than proof. Their purpose is to test whether the same structure clarifies problems that differ substantially in domain, stakes, and degree of experimental control.

Appendix B extends this comparison with a compact classification of cases by identified conjugacy and responsive embedding, including harder boundary cases not developed in the main text.

6.1 Portfolio concentration risk

For a portfolio risk manager, different structural accounts of concentration can imply different stress scenarios and different hedges. The case is strongest as an example of decision-relative problem formulation and structural conjugacy; identified conjugacy may be weak when diagnostic stress events are rare or when the manager's own hedge is too small to alter the market.

6.2 Making money in adaptive financial markets

A trader may attribute a return opportunity to temporary mispricing, persistent risk compensation, institutional constraints, behavioral regularities, or informational advantage. These accounts imply different expectations about durability. The Adaptive Markets Hypothesis already emphasizes competition and adaptation in changing financial environments [13]. The conjugacy view adds a focused mechanism: useful attribution can support useful prediction, and successful exploitation can then help alter the opportunity being exploited.

$$A \rightarrow P \rightarrow \text{trade} \rightarrow S' \rightarrow P_{\text{old}} \downarrow$$

A deteriorating strategy can therefore mean very different things: the pattern may have been spurious, the relevant structure may have changed independently, or exploitation may have weakened the signal. The responsive embedding generates the possibility of decay; the conjugacy view contributes the three-way diagnostic interpretation.

This mechanism applies most directly when the opportunity is competitively exploitable and exploitation itself affects the relation generating the signal. It is not a general claim that every source of excess return must disappear under use.

6.3 Adaptive clinical trials

Clinical trials provide a relatively controlled setting for the conjugate core. Competing accounts of treatment response can imply different outcomes under prospectively specified treatments, and observed outcomes can revise those accounts [11]. Adaptive features may change allocation or stopping rules as evidence accumulates, but adaptation by the trial does not imply that the underlying disease system is itself responsive.

The useful contrast is experimental access: well-designed randomization, enrichment, or stress conditions can create diagnostic variation that helps distinguish competing accounts. Adaptive design itself does not automatically increase identification and can involve efficiency or power trade-offs.

6.4 Public policy during COVID-19

COVID-19 policy illustrates a much less controlled environment. Governments acted while disease transmission, behavior, treatment capacity, immunity, variants, mobility, economic conditions, and public responses changed. OECD's synthesis of early government evaluations emphasizes the role of evaluation in learning what was working, what was not, what could work, and for whom [12].

The conjugacy view provides one reason policy learning can remain provisional: evidence needed to distinguish among competing accounts may arrive while the relevant behavioral or epidemiological structure is also changing. COVID is therefore an illustration of a possible regime in which λ_I is low relative to λ_R , not an empirical estimate of those rates from the OECD source.

6.5 A negative control: carrying an umbrella

Suppose a forecast gives a high probability of rain tomorrow and the decision is whether to carry an umbrella. For this task, the prediction is usually enough. The atmospheric mechanism may be scientifically interesting, but it generally does not change the decision. Carrying the umbrella does not alter the weather or create useful diagnostic evidence about atmospheric dynamics.

$$P \rightarrow D$$

Prediction–attribution conjugacy adds little. That is important. A useful framework should not make every problem look like an instance of itself.

7. Relation to Existing Work

Prediction and attribution sit near several established research traditions. The present framework does not replace those traditions. It organizes a particular relationship among problems they study from different directions.

A natural starting point is the distinction between prediction and explanation. Shmueli distinguishes explanatory and predictive modeling goals [1], while Hofman and colleagues argue that explanation and prediction can complement rather than simply compete [2]. The present paper asks a narrower question: when prediction and decision-relevant attribution are both involved in the same adaptive decision problem, how do they relate to one another?

Causal inference and policy analysis distinguish observational prediction from intervention-oriented questions. Kleinberg and colleagues identify an important class of policy problems in which predictive inference has an independent role [3]. Causal-prediction work studies predictions under hypothetical interventions [6], while invariant prediction uses predictive stability across environments as information about causal structure [7].

Adaptive control and system identification already contain much of the outer loop considered here: model, prediction, action, observation, and update. Dual control makes the informational role of action especially explicit [8]. The present paper focuses more narrowly on the inner relation between decision-relevant attribution and prediction.

Bayesian experimental design provides a close precedent for choosing observations or interventions that discriminate among competing models [14]. Decision-focused learning likewise emphasizes that predictive accuracy and decision quality need not coincide [15]. Performative prediction shows that predictions used for decisions can change the future distribution the predictor faces [9].

Two literatures use terminology particularly close to the present paper. Efron distinguishes prediction, estimation, and attribution in statistical inference [4]. Rayl and Sinha explicitly integrate attribution-based

and pure-prediction approaches and document trade-offs between them [5]. Tikhonov and DeDeo study prediction, explanation, and control as related but distinct abilities in dynamic environments [10].

Taken together, these literatures establish many ingredients used here. The present framework takes a different organizing view: it places the reciprocal relation

$$A_D \leftrightarrow P_D$$

at the center and asks what becomes easier to understand when that relation is embedded in consequential decisions. The division of labor is important: Bayesian discrimination and related methods formalize parts of the conjugate core, while performativity, dual control, and other adaptive literatures formalize parts of the surrounding dynamics. The present framework's role is to organize these pieces around decision-relevant attribution and diagnostic prediction, not to replace them.

8. Implications and Limits

8.1 Monitoring the pair, not only the predictor

Model monitoring commonly asks whether predictive performance remains acceptable. In a conjugate problem, average performance under routine conditions may be insufficient.

$$P_t \text{ stable under non-diagnostic conditions} \not\Rightarrow A_t \text{ stable}$$

$$A_t \text{ stable} \not\Rightarrow P_t \text{ stable across all relevant conditions}$$

If competing attributions differ only under stress, intervention, subgroup, or shifted environments, stable average accuracy provides little evidence about which attribution remains valid. Attributional validity is not directly observed; it is assessed through predictions under conditions chosen because the competing accounts make different claims. Pair-level monitoring should therefore preserve or create such diagnostic contrasts—for example through holdouts, randomized exposure where feasible, stress scenarios, or subgroup and shift slices. The practical question becomes: Is the prediction still useful for the same reason, and have we tested that reason where it could fail?

8.2 Understanding can improve without becoming final

At any point, several decision-relevant attributions may remain compatible with available evidence. New evidence can reduce this uncertainty, but the relevant attributional structure may also change while that evidence is being accumulated. Understanding can therefore improve without converging to a permanently fixed representation.

The identification comparison offers one way to organize this possibility. Where diagnostic learning substantially outruns structural change, understanding may stabilize. Where the two proceed on similar timescales, understanding can remain persistently provisional.

A related extension may apply to life choices: a decision can be reasonable given the information and self-understanding available at one time even if later choices differ because the decision itself changed opportunities, preferences, or the decision maker. This is an extension of the framework rather than a core result.

8.3 The current AI landscape as an application domain

The current AI landscape is not supporting evidence for the framework. It is a domain in which the framework’s diagnostic questions may be useful: which structural accounts support a prediction, where do competing accounts diverge, and how might deployment change the relation being modeled?

Future observations can show whether those questions remain useful as the domain changes.

8.4 Limits of the framework

The framework should not be applied merely because a problem contains prediction, explanation, feedback, or adaptation. Its broad problem population is defined independently by consequential choice under structural uncertainty, later observation, and possible revision. Structural conjugacy asks whether competing decision-relevant accounts imply different consequences somewhere; identified conjugacy asks whether feasible diagnostic variation can actually distinguish them in time for the decision.

Some problems lie outside this population. Others lie inside it but remain non-identified because the relevant diagnostic condition cannot be observed or created before action is required. Still others may be better understood through different frameworks entirely.

Appendix C collects several exploratory extensions—including the moving-frontier, information-bound, identification-race, and continuous-time ideas. These are not required by the main argument, but are included to show where the framework might be developed further.

9. Conclusion

This paper began with a practical problem. A decision maker facing an uncertain system often needs more than a forecast of what will happen. The decision may also depend on understanding which parts, relations, or mechanisms within that system are responsible for the outcomes that matter.

In such problems, prediction and attribution can become linked in more than one way. Attribution can guide what should be predicted; more strongly, competing attributions can imply different outcomes under the same informative condition, allowing predictive evidence to bear on their credibility.

$$A_D \leftrightarrow P_D$$

We describe the latter reciprocal relation as prediction–attribution conjugacy. The conjugacy is neither an exact inversion nor a universal trade-off. It is a decision-relative diagnostic relation between structural accounts and prospective outcomes.

When that conjugate core is embedded in a responsive system, action may alter the structure on which the original prediction and attribution depended. Conjugacy does not create the feedback; it helps interpret it. This is why successful intervention can weaken an old predictive signal, why routine predictive stability may fail to validate the underlying attribution, and why diagnostic learning can race against structural change.

These explanations do not replace existing economic, causal, statistical, psychological, or control-theoretic accounts. They provide another vantage point whose strongest value is diagnostic: identifying which structural distinctions matter to the decision, what evidence could distinguish them, and how to interpret changes after action.

The value of the framework depends less on whether it provides an exclusive explanation than on whether it makes important problems easier to see, distinguishes cases that should be treated differently, and improves the next question a decision maker asks.

What understanding supports the prediction, how can that understanding be tested, and what happens to both once we act on them?

AI-Use Disclosure

AI tools were used in the development of this working paper for exploratory discussion, literature discovery, drafting assistance, and editorial revision. The author selected the research question, developed and directed the argument, evaluated the resulting claims and structure, and takes responsibility for the final content.

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Appendix A. Formal Definitions and Minimal Model

This appendix states the core ideas more formally. The purpose is clarification, not to impose a stronger mathematical structure than the argument requires.

A.1 Decision environment

Let S denote the broader system, D a decision objective, and

$$S_D = \pi_D(S)$$

the decision-relevant representation. Let $u \in U_D$ denote an available action and Y a consequential outcome. The decision problem is defined over (D, S_D, U_D, Y) .

A.2 Prediction

A predictive mapping associates the decision-relevant system and a relevant action or condition with a prospective distribution of outcomes:

$$P_D : (S_D, u) \mapsto P(Y | S_D, u)$$

Prediction need not identify why the outcome occurs.

A.3 Attribution

Let A_D denote a family of attributional hypotheses about the decision-relevant system. Each A_i is a candidate account of which components, relations, or mechanisms within S_D matter to consequential differences in outcomes under relevant alternatives.

An attribution can exist even when its prospective implications are not yet distinct or empirically identifiable. Predictive distinguishability is therefore a property to be assessed, not part of the attribution definition.

A.4 Attributional equivalence

For decision objective D , define

$$A_i \sim_D A_j$$

when the distinction between A_i and A_j does not change any relevant decision. The equivalence class $[A]_D$ therefore contains attributions that may differ scientifically or descriptively but are indistinguishable for the decision at hand.

A.5 Predictive distinguishability

Two attributional classes are predictively distinguishable if, for a fixed outcome of interest and at least one common relevant condition or action u , they imply different prospective outcome distributions:

$$P_D(Y | u, A_i) \neq P_D(Y | u, A_j)$$

Here u is shorthand for a contemplated or implemented action or diagnostic condition; when causal semantics are warranted, it should be interpreted interventionally. Predictive distinguishability is not assumed by the attribution definition; it is an additional property used in the conjugacy test.

A.6 Prediction–attribution conjugacy

For a decision task D , prediction and attribution exhibit structural conjugacy when at least some decision-relevant attributional distinctions generate different prospective predictions for the same outcome under a common informative condition. They exhibit identified conjugacy when available observations or interventions are sufficiently diagnostic to discriminate among those competing attributions before relevant structural change invalidates the comparison.

$$A_D \leftrightarrow P_D$$

The notation represents reciprocal prospective implication and diagnostic evidence. It does not imply that prediction and attribution are inverses, equally accurate, or sufficient for unique causal identification.

A.7 Learning loop and responsive embedding

$$u_t = h_D(A_t, P_t)$$

$$S_{D,t+1} = T(S_{D,t}, u_t, \varepsilon_{t+1})$$

Observed outcomes may then contribute to revision from (A_t, P_t) to (A_{t+1}, P_{t+1}) ; this is the learning loop. Responsiveness of the system is an additional condition: action must materially change the relevant outcome distribution or attributional structure, whether mechanically or through reaction by other actors. Endogenous validity loss and success-induced signal attenuation require this responsive embedding.

A.8 Predictive sufficiency and decision sufficiency

Predictive sufficiency for a particular target does not imply decision sufficiency when appropriate action depends on additional consequential outcomes, conditions, or structural distinctions that the target does not encode.

$$\text{predictive sufficiency for one target} \not\Rightarrow \text{decision sufficiency}$$

A.9 Success-induced signal attenuation

Let $X \subset S_D$ materially contribute to Y , supply predictive information about Y , and be identified as actionable. If action changes the distribution of X or the relation between X and Y , then it is possible that

$$I(X; Y)_{\text{post}} < I(X; Y)_{\text{pre}}$$

and a predictor relying on the pre-intervention relation can deteriorate. This deterioration does not by itself imply that the prior attribution or intervention was unsuccessful.

A.10 Pair-level validity

Let $V_P(t)$ denote predictive validity under the conditions being monitored, and let $V_A(t)$ denote the evidentiary credibility of the attributional account relative to its competitors under diagnostic conditions.

$$V_P(t + 1) \approx V_P(t) \text{ under non-diagnostic conditions} \not\approx V_A(t + 1) \approx V_A(t)$$

Attributional validity is not independently observable; it is assessed through diagnostic predictions, interventions, or other evidence under conditions where competing accounts diverge. Average predictive stability under routine conditions therefore cannot by itself establish attributional stability.

A.11 The identification race

$$\rho = \lambda_I / \lambda_R$$

Here λ_I is the conceptual rate at which evidence reduces uncertainty among competing decision-relevant attributions and λ_R is the rate at which the relevant attributional structure changes enough to invalidate the comparison. What counts as invalidating change depends on the diagnostic comparison itself, so the ratio is heuristic rather than a pair of fully independent measured rates.

A.12 A simple diagnostic toy model

Suppose a binary outcome Y is predicted from a signal X . Two attributions fit the pre-intervention data equally well. A_I says X is part of the mechanism generating Y ; A_2 says X is only a proxy for an unobserved driver Z . Before intervention, both models yield $P(Y = 1 | X = 1) = 0.60$.

A decision maker then intervenes to reduce X in a treated group while preserving an otherwise comparable untreated group. Under A_I , the treated event rate is predicted to fall to 0.20 while the untreated group should remain near 0.60. Under A_2 , reducing X alone should not materially change the event rate because Z remains unchanged. If the treated rate is 0.25 and the untreated rate remains near 0.60 under conditions controlled well enough for the task, the observation shifts support toward A_I .

The comparison also makes the three-way diagnosis concrete. If the historical signal deteriorates similarly in treated and untreated groups, the original relation may have been spurious or the structure may have drifted independently. If deterioration is concentrated in the treated group and is close to the change predicted by A_I , success-induced attenuation is more plausible. Here the mechanism need not have failed or changed: the intervention may simply have compressed the distribution of X , leaving the attribution intact while weakening the old predictive signal. An accuracy-only monitor records deterioration; pair-level monitoring asks which pattern of deterioration occurred.

Appendix B. Extended Cases and Classification

This appendix extends the case analysis. The cases were selected to clarify boundaries, including both in-class problems and out-of-class controls.

The classification uses two independent dimensions: whether identified conjugacy is available in practice, and whether a responsive embedding is present. Structural conjugacy may exist even when diagnostic variation is unavailable; the identified dimension therefore carries the empirical burden.

Table B1. Identified conjugacy and responsive embedding

	Responsive embedding absent	Responsive embedding present
Identified conjugacy absent	One-off structural decision: competing accounts matter for action, but no feasible diagnostic condition is available before the choice. Weather → umbrella remains an out-of-class control because attribution is not decision-relevant.	Responsive policy or platform setting with structural uncertainty but no feasible diagnostic contrast before the system changes; reactive recommender is an out-of-class control when attribution is operationally irrelevant.
Identified conjugacy present	Adaptive clinical trial or stationary reliability diagnosis: competing accounts make testable prospective claims, while the underlying mechanism need not respond to the diagnostic act.	Adaptive finance or performative credit: competing accounts can be diagnostically separated and action can alter the relevant opportunity or population.

The purpose of the matrix is not to rank domains. It is to show that diagnostic identifiability and responsiveness are separate properties, and that the same domain can move between cells depending on the decision architecture.

B.1 What the extended cases add

The extended cases reinforce three distinctions:

$$\begin{aligned} \text{adaptive feedback} &\neq \text{prediction-attribution conjugacy} \\ \text{structural conjugacy} &\neq \text{identified conjugacy} \\ \text{domain} &\neq \text{classification} \end{aligned}$$

The same domain can contain identified, non-identified, responsive, and non-responsive problems depending on how prediction, attribution, decision, diagnostic access, and feedback are related. The framework is therefore best understood as a classification of problem structures, not of fields.

Appendix C. Further Conjectures and Open Directions

This appendix collects a small set of ideas that arise naturally from the framework but are not established by the main paper.

C.1 A moving attainable frontier

$$F_t(D, I_t) = \{(q_A, q_P)\}$$

No universal negative trade-off is assumed. Better attribution may improve prediction; successful intervention may weaken an old predictive relation; and in other settings the two may change largely independently. The open question is whether conjugate systems exhibit useful common structure in the boundary and movement of this attainable set.

C.2 Information and identifiability limits

If decision-relevant states are observationally indistinguishable under current information but imply different appropriate actions, no analytical method can recover that distinction from the existing evidence alone. The

joint resolution of prediction and attribution may therefore be bounded by the finest decision-relevant distinctions recoverable from available information, measurement, variation, and intervention.

C.3 The identification race as a dynamic limit

$$\rho = \lambda_I / \lambda_R$$

Stable understanding may require not only sufficient information but a rate of identification that exceeds the rate at which the relevant attributional structure changes.

C.4 Continuous-time limit

As $\Delta t \rightarrow 0$, attribution, prediction, decision, and system evolution may become continuously co-evolving rather than cleanly sequential. A future formalization could investigate coupled dynamics such as $A(t)$, $P(t)$, $D(t)$, and $S(t)$, without assuming that every conjugate problem admits the same differential-equation representation.